

Instructor: Date:

Learner: Regional Branch:

Program: Center:

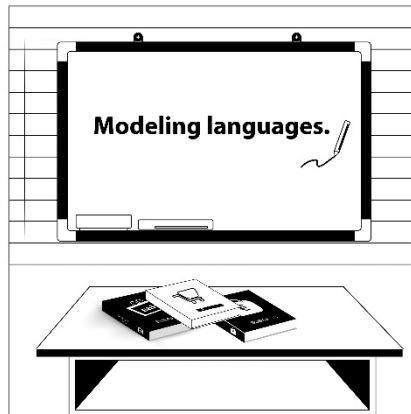
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Let's Start!

How to Build Software



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can talk about modeling languages.
- I can exchange information about modeling languages.
- I can give examples of modeling languages and talk briefly about them.

Ask your partner:

- How do you think programmers create software?
- Do you think that the software programs we use today are different from the ones we used 10 years ago? What changed?





Let's Explore!

Kyle checks out* his knowledge about modeling languages.

*To "check out" means to inspect something.



Instructions: Read the text below.

- Kyle:** Hey, Mandy! There is something I need to summarize for my class workshop. Can you help me?
- Mandy:** Sure, Kyle! I guess it's related to software.
- Kyle:** Yes, it is! I need to describe modeling languages and talk about them. I watched a documentary and it showed the origin of modeling languages.
- Mandy:** Ok, tell me about the documentary so I have a better idea of what you need.
- Kyle:** Well, I know that modern software projects are complex, in other words, a software is not a program itself but a series of systems or modules. Because they are so large, there are easier ways to design those software projects.
- Mandy:** That's right! And that's when programmers modeled software projects using modeling languages. With them, it is possible to map out the structure of those programs like the components and the relationships they hold.
- Kyle:** Oh I see!! There are significant software projects that use the modeling languages to make their creation easier. I understand that the Universal Modeling Language is the most used in programming.
- Mandy:** Exactly! It is the fastest and easiest way to shape the structure of a program. It describes software in three ways: functions seen by the user; objects, attributes, and relationships; and system behavior.
- Kyle:** Ok, I guess it's necessary to use a modeling language to structure a software before people code the program, right?
- Mandy:** That's right! That's how programmers interact easier with programs. It makes creation easier. Just take into account that UML uses a variety of diagrams to shape different processes.

**Let's Practice!****One Example of Modeling Languages**

Instructions: Fill in the gaps to correctly complete the sentences. Use: Modeling Language, software projects, and Modern.

1. There are different modeling software, one example is The Universal _____.
2. There is a special characteristic to remember related to programming processes and _____.
3. _____ software projects are complex because of the series of systems and modules behind them.

**Let's Practice!**

Yesterday, Paula and I studied for our software exam!

Instructions: Select the correct answer.

1. Yesterday, Paula and I studied for our software exam and we _____ some characteristics of software programming.
Hint: Past tense
 - a. summarized
 - b. summarize
 - c. summarizing
 - d. summarizes
2. Then, we _____ the Universal Modeling Language.
Hint: Past tense
 - a. describing
 - b. described
 - c. describes
 - d. describe
3. We learned about how the programmers _____ software projects applied modeling languages.
Hint: Past tense
 - a. model
 - b. to model
 - c. modeling
 - d. modeled





Extension Activity

Let's summarize modeling languages.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about modeling languages.
- Make a list of 10 characteristics of modeling languages.
- Share your list with the class.

Self-Assessment

Yes

No

1. I look for more information about modeling languages.

☐ ☐

2. I list characteristics of modeling languages.

☐ ☐

3. I share my work with the class.

☐ ☐


Target Words

- **significant** (adjective): a representative quantity
e.g. There are significant software projects that use the modeling languages to make their creation easier
- **be related to** (expression): to have a connection or be linked to something
e.g. I guess it's related to software.
- **series** (noun): a number of things that share characteristics
e.g. a software is not a program itself but a series of systems or modules.
- **map out** (phrasal verb): to plan or design something with detail
e.g. With them, it is possible to map out the structure of those programs...
- **summarize** (verb): to express an idea in a shorter and more synthesized way
e.g. There is something I need to summarize for my class workshop.
- **shape** (verb): to give form
e.g. It is the fastest and easiest way to shape the structure of a program.



Learning Points

Adjectives and Nouns to Talk about Modeling Languages

There are many words that people use to talk about specific or more specialized topics.

When talking about modeling languages, we can use some words that are special to this topic.

Nouns like **software**, **systems**, **modules**, are characteristic when talking about computers and related things.

Adjectives like **significant**, **modern**, **large**, or **complex**, give attributes to those nouns.

Examples

- *I know that modern software projects are complex.*
- *Because they are so large, there are easier ways to design those software projects.*
- *There are significant software projects that use the modeling languages to make their creation easier.*

Action Verbs in Simple Present and Simple past to Talk Modeling Languages

Action verbs talk about actions people perform.

Action verbs describe a wide variety of actions people perform every day. In this case, there are some action verbs we can use to talk about modeling languages.

Verbs like **model**, **code**, **interact**, **summarize**, **describe**, and **show**, have different meanings based on different contexts. However, if we talk about software and related topics, these verbs change their meaning. For example, the verb **model** changes its meaning if we talk about fashion.

Those verbs can be used in present or in past. Because those are regular verbs, to use them in past you only need to add **-ed** at the end of the verb.

Let's take a look at the examples below.

Examples

- *There is something I need to summarize for my class workshop.*
- *I need to describe modeling languages and talk about them.*
- *And that's when programmers modeled software projects using modeling languages.*

